

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Second Semester B.Tech Degree Examination June 2022 (2019 scheme)

Course Code: PHT100

Course Name: ENGINEERING PHYSICS A

(2019 -Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks

- | | | Marks |
|----|--|-------|
| 1 | Comment the statement “smaller the damping, larger will be the relaxation time and greater the quality factor”. | (3) |
| 2 | State the laws of vibrations of a stretched string. | (3) |
| 3 | Explain the formation of different colours in thin films with example. | (3) |
| 4 | What is diffraction? Compare diffraction pattern with interference pattern. | (3) |
| 5 | Write the expression for energy of a particle trapped in one dimensional box and find the energy of first excited state. | (3) |
| 6 | What is meant by quantum dot and quantum wire? | (3) |
| 7 | State Ampere’s circuital theorem. Give its mathematical statement. | (3) |
| 8 | Define Divergence of a vector function. Give its physical significance. | (3) |
| 9 | Write a note on high temperature superconductors with two examples. | (3) |
| 10 | Explain the working of a solar cell. | (3) |

PART B

Answer one full question from each module, each question carries 14 marks.

MODULE 1

- 11 a) Discuss the conditions of over damped, critically damped and under damped cases of a damped harmonic oscillator by writing down the differential equation and its solution. With the help of displacement - time graph show that how damping affects the amplitude of the oscillator. (10)
- b) Show that LCR circuit acts as an electrical analogue of mechanical oscillator. (4)
- 12 a) Considering the transverse vibrations of a stretched string derive the one dimensional wave equation and show that velocity of the wave $v = \sqrt{T/m}$. (10)

- b) The displacement of a wave is represented by $y = 0.25 \times 10^{-3} \sin(0.025x - 500t)$, where x is measured in metres and t in secs. Deduce amplitude, time period, velocity and wavelength? (4)

MODULE 2

- 13 a) Explain the construction and band formation in Newton's ring apparatus. Derive an equation to find the wavelength of incident monochromatic light. (10)
- b) What is an antireflection coating? Find the minimum thickness of antireflection coating (magnesium fluoride of refractive index **1.42**) to be deposited on a glass plate of refractive index **1.5**, for light of wave length **5800 Å**. (4)
- 14 a) Explain resolving power of a grating? Explain the formation of spectra by a plane transmission grating by deriving grating equation. (10)
- b) Consider a plane transmission grating having **6000** lines/cm. Find the angular separation between first and second order maxima when a light of wavelength **5500 Å** incident normally on it. (4)

MODULE 3

- 15 a) Derive time dependent Schrodinger equation for a particle moving in a potential $V(x)$. Hence derive time independent Schrodinger equation. (10)
- b) State and explain Heisenberg's uncertainty principle. Explain natural line broadening mechanism on the basis of this principle. (4)
- 16 a) What are nanomaterials? Discuss the mechanical, optical and electrical properties of nano materials. (8)
- b) Give any six applications of nanomaterials. (6)

MODULE 4

- 17 a) Explain the terms (i) Magnetization (ii) Magnetic permeability (iii) Relative permeability and (iv) Susceptibility. Derive the relation between magnetic susceptibility and relative permeability. (10)
- b) The magnetic flux through a closed circuit with resistance **2.5 Ω** varies with time obeying the equation $\phi = 5t^2 + 2t + 6$. What will be the induced emf and current in it at time = **5** seconds. (4)

- 18 a) Starting from the field equations in free space, derive the velocity of electromagnetic waves in free space. How did Maxwell show that light is electromagnetic in nature? (10)
- b) State and explain Poynting's theorem. (4)

MODULE 5

- 19 a) What is critical temperature and critical field? Write the relation between these two. Explain the conductivity of superconductors using BCS theory. (10)
- b) State and explain Meissner effect. Show that superconductors are perfect diamagnets. (4)
- 20 a) Describe fibre optic communication system with a block diagram. List four advantages of fibre optic communication. (10)
- b) The numerical aperture of an optic fibre is **0.38**. If the difference in the refractive indices of the material of its core and the cladding is **0.05**, calculate the refractive index of material of the core. (4)
